

CSA0904 CO2 AT1

Q1.Student Registration System

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (VASAMSETTI HARSHITHA). The main content area is titled "CO2-AT1-Scenario Based Programs" and contains a list of questions on the left and a detailed question on the right. The question is "Q1 Student Registration System" (10 marks). The question text describes a university system and asks for three tasks: (a) Enumerate three Collection interfaces, (b) Design a Collection structure, and (c) Write a Java program using HashSet and HashMap with Generics. The "Your Code" section shows a partial Java solution for the main method, including HashSet and HashMap declarations and student additions. The "Input" section is empty, and the "Output" section shows the expected output for the given input.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         HashSet<Student> students= new HashSet<>();
5         Student s1=new Student(101,"Harshi");
6         Student s2=new Student(102,"Aamna");
7         students.add(s1);
8         students.add(s2);
9         HashMap<Student,ArrayList<String>> map=new HashMap<>();
10        ArrayList<String> c1=new ArrayList<>();
11        c1.add("java");
```

Input

stdin input

Output

```
Student:101 - Harshi
Courses:[java, python]
Student:102 - Aamna
Courses:[java, python]
```

This screenshot shows the same SIMATS Online Programming Lab interface, but with a complete Java solution for the "Q1 Student Registration System" question. The "Your Code" section now includes the full implementation of the Student class and the main method, which uses HashSet to store unique students and HashMap to map students to their enrolled courses. The "Input" section is empty, and the "Output" section shows the expected output for the given input.

QUESTIONS

- Q5 Online Voting System 10 marks

5/5 answered

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         HashSet<Student> students= new HashSet<>();
5         Student s1=new Student(101,"Harshi");
6         Student s2=new Student(102,"Aamna");
7         students.add(s1);
8         students.add(s2);
9         HashMap<Student,ArrayList<String>> map=new HashMap<>();
10        ArrayList<String> c1=new ArrayList<>();
11        c1.add("java");
12        c1.add("python");
13        ArrayList<String> c2=new ArrayList<>();
14        c2.add("java");
15        c2.add("python");
16        map.put(s1,c1);
17        map.put(s2,c2);
18        Iterator<Student> it=students.iterator();
19        while(it.hasNext()){
20            Student s=it.next();
21            System.out.println("Student:"+s);
22            System.out.println("Courses:"+map.get(s));
23            System.out.println();
24        }
25    }
26 }
27 class Student{
28     int id;
29     String name;
```

Input

stdin input

Output

```
Student:101 - Harshi
Courses:[java, python]
Student:102 - Aamna
Courses:[java, python]
```

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```
24 }
25 }
26 }
27 class Student{
28     int id;
29     String name;
30     Student(int id,String name){
31         this.id=id;
32         this.name=name;
33     }
34     public int hashCode(){
35         return id;
36     }
37     public boolean equals(Object obj){
38         Student s=(Student) obj;
39         return this.id==s.id;
40     }
41     public String toString(){
42         return id+" - "+name;
43     }
44 }
```

Run Save

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32°C Mostly cloudy

Search

ENG IN

10:42 06-08-2026

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Q2.E-Commerce Cart System

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QUESTIONS

Q1 Student Registration System 10 marks

Q2 E-Commerce Cart System 10 marks

Q3 Library Management System 10 marks

Q4 Employee Data Processing 10 marks

Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         ArrayList<String> cart=new ArrayList<>();
5         cart.add("Laptop");
6         cart.add("Keyboard");
7         System.out.println("Cart items:");
8         Iterator<String> it=cart.iterator();
9         while(it.hasNext()){
10             System.out.println(it.next());
11         }
12     }
13 }
```

Input

stdin input

Output

Cart items:
Laptop
Keyboard

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Search

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10:45 06-08-2026

Q3.Library Management System

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=747`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "VASAMSETTI HARSHITHA" with ID "192372385RA". The page is categorized under "JAVA PROGRAMMING".

On the left, a sidebar lists five questions (Q1 to Q5) with their respective marks (10 marks each). Q3, "Library Management System", is currently selected. Below the list, it indicates "5/5 answered".

The main content area displays the question text: "Suppose that a library stores books with unique ISBN numbers and needs fast retrieval. (a) Name three Map implementations in Java. (b) Design a schema-like representation using Map for storing ISBN and book details. (c) Write a Java program using HashMap to: • Insert book details • Retrieve a book • Display all books".

Below the question, there is a "Your Code" section with a text editor showing the following Java code:

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         HashMap<String,String> library=new HashMap<>();
5         library.put("101","java");
6         library.put("102","python");
7         System.out.println("Book with isbn 102:"+library.get("102"));
8         System.out.println("Library Books:");
9         for(Map.Entry<String,String> book:library.entrySet()){
10             System.out.println(book.getKey());
11             System.out.println(book.getKey()+" : "+book.getValue());
12         }
13     }
14 }
15 }
```

To the right of the code editor is an "Input" field with the text "stdin input" and an "Output" field showing the program's output:

```
Book with isbn 102:python
Library Books:
101 : java
102 : python
```

The bottom of the screen shows a Windows taskbar with various application icons, a search bar, and system tray information including temperature (32°C), weather (Mostly cloudy), and time (10:47, 06-08-2026).

Q4.Employee Data Processing

The screenshot shows the SIMATS Online Programming Lab interface for Q4: Employee Data Processing. The browser address bar displays the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=747`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "VASAMSETTI HARSHITHA" with ID "192372385RA". The page is categorized under "JAVA PROGRAMMING".

On the left, a sidebar lists five questions (Q1 to Q5) with their respective marks (10 marks each). Q4, "Employee Data Processing", is currently selected. Below the list, it indicates "5/5 answered".

The main content area displays the question text: "Suppose that a company processes employee records and filters employees based on salary. (a) List three advantages of Generics in Collections. (b) Represent how employee data is stored using List with Generics. (c) Write a Java program using Iterator to: • Store employee objects • Display employees with salary > 50,000".

Below the question, there is a "Your Code" section with a text editor showing the following Java code:

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         ArrayList<Employee> list=new ArrayList<Employee>();
5         list.add(new Employee(101,"Rahul",60000));
6         list.add(new Employee(102,"Raghu",55000));
7         System.out.println("Employees with salary>50000:");
8         Iterator<Employee> it=list.iterator();
9         while(it.hasNext()){
10             Employee e=it.next();
11             if(e.salary>50000){
12                 System.out.println(e.id+" "+e.name+" "+e.salary);
13             }
14         }
15     }
16 }
```

To the right of the code editor is an "Input" field with the text "stdin input" and an "Output" field showing the program's output:

```
Employees with salary>50000:
101 Rahul 60000.0
102 Raghu 55000.0
```

The bottom of the screen shows a Windows taskbar with various application icons, a search bar, and system tray information including temperature (32°C), weather (Mostly cloudy), and time (10:48, 06-08-2026).

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10 marks
Q4 Employee Data Processing 10 marks
Q5 Online Voting System 10 marks
5/5 answered

Your Code

```

1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         ArrayList<Employee> list=new ArrayList<Employee>();
5         list.add(new Employee(101,"Rahul",60000));
6         list.add(new Employee(102,"Raghu",55000));
7         System.out.println("Employees with salary>50000:");
8         Iterator<Employee> it=list.iterator();
9         while(it.hasNext()){
10             Employee e=it.next();
11             if(e.salary>50000){
12                 System.out.println(e.id+" "+e.name+" "+e.salary);
13             }
14         }
15     }
16 }
17 class Employee{
18     int id;
19     String name;
20     double salary;
21     Employee(int id,String name,double salary){
22         this.id=id;
23         this.name=name;
24         this.salary=salary;
25     }
26 }

```

Input
stdin input
Output
Employees with salary>50000:
101 Rahul 60000.0
102 Raghu 55000.0

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Search
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10:48 06-08-2026

Q5.Online Voting System

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10 marks
Q1 Student Registration System 10 marks
Q2 E-Commerce Cart System 10 marks
Q3 Library Management System 10 marks
Q4 Employee Data Processing 10 marks
Q5 Online Voting System 10 marks
5/5 answered

Q5 Online Voting System

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```

1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         HashSet<String> voters=new HashSet<>();
5         HashMap<String,Integer> votes=new HashMap<>();
6         voters.add("V101");
7         votes.put("Alice",1);
8         voters.add("V102");
9         votes.put("Bob",1);
10        if(voters.add("V101")){
11            votes.put("Alice",1);
12        }else{
13            System.out.println("Duplicate vote");
14        }
15        System.out.println("Results:");
16        System.out.println("Alice="+votes.get("Alice"));

```

Input
stdin input
Output
Duplicate vote
results:
Alice=1
Bob=1

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Search
ENG IN
10:49 06-08-2026